

Data Mining (EECS6412)

Frequent Pattern and Association Rule Mining

Aijun An

(Modified from Jiawei Han's slides)

Outline

- ▶ Basic concepts of association rule learning
- ▶ Apriori algorithm
- ▶ FP-Growth algorithm
- ▶ Finding interesting rules.

Why Mining Association Rules?

- ▶ Objective:

- ▶ Finding interesting co-occurring items (or objects, events) in a given data set.

- ▶ Examples:

- ▶ Given a dataset of transactions, each transaction is a list of items (purchased by a customer during a visit), you may find:
 - ▶ $\{\text{bread, butter}\} \rightarrow \{\text{milk}\}$ [support=2%, confidence=60%]
 - ▶ From a student database, you may find
 - ▶ $\text{major}(x, \text{"CS"}) \wedge \text{gpa}(x, \text{"A"}) \rightarrow \text{has_taken}(x, \text{"DB"})$ [1%, 75%]

Why Mining Association Rules?

(Cont'd)

- ▶ Popular application: *Basket data analysis*
 - ▶ Place items frequently bought together close to each other to increase sales of these items.
 - ▶ ? $\rightarrow$ *iPad*
 - ▶ What the store should do to boost sales of the particular product, i.e., *iPad*
 - ▶ *iPad* $\rightarrow$?
 - ▶ What other products should the store stock up?
 - ▶ Make recommendations

What Kind of Datasets?

Transactional dataset TDS

TID	Items
100	f, a, c, d, g, i, m, p
200	a, b, c, f, l, m, o
300	b, f, h, j, o
400	b, c, k, s, p
500	a, f, c, e, l, p, m, n

- ▶ Itemset: a set of items
- ▶ A **transaction** is a tuple (tid, X)
 - ▶ Transaction ID tid
 - ▶ Itemset X
- ▶ A **transactional dataset** is a set of transactions
 - ▶ In many cases, a transaction database can be treated as a set of itemsets (ignore TIDs)
- ▶ Association rule from TDS (relates two itemsets):
 - ▶ $\{a, c, m\} \rightarrow \{l\}$ [support=40%, confidence=66.7%]

What Kind of Datasets? (*Contd*)

Tabular data

Day	Outlook	Temp	Humid	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- ▶ An attribute-value pair
 - ▶ E.g., Outlook = sunny
- ▶ Record: a set of attribute-value pairs
- ▶ Table: a set of records

- ▶ Association rule from Table (relates two sets of attribute-value pairs):
(Outlook=sunny) $\wedge$ (Temp=hot) $\rightarrow$ (PlayTennis=No) [support=14%, confidence=100%]

Definition of Association Rule

- ▶ An association rule is of the form:

$X \rightarrow Y$ [support, confidence]
where antecedent consequent

- ▶ $X \subset I$, $Y \subset I$, $X \cap Y = \emptyset$ and I is a **set** of items (or attribute-value pairs).
- ▶ **support**: probability that a transaction (or a record) contains X and Y , i.e.,

$$\text{support}(X \rightarrow Y) = P(X \cup Y)$$

- ▶ **confidence**: conditional probability that a transaction (or a record) having X also contains Y , i.e.,

$$\text{confidence}(X \rightarrow Y) = P(Y|X)$$

- ▶ A rule associates one set of items (or attribute-value pairs) with another set of items (or attribute-value pairs)

Support and Confidence: Example

$$\text{support}(X \rightarrow Y) = P(X \cup Y)$$

$$\text{confidence}(X \rightarrow Y) = P(Y|X)$$

Transaction ID	Items Bought
2000	A,B,C
1000	A,C
4000	A,D
5000	B,E,F

Relative frequency is used to estimate the probability.

- ▶ $\{A\} \rightarrow \{C\}$ (50%, 66.7%)
- ▶ $\{C\} \rightarrow \{A\}$ (50%, 100%)
- ▶ $\{A, C\} \rightarrow \{B\}$ (25%, 50%)
- ▶ $\{A, B\} \rightarrow \{E\}$ (0%, 0%)

Mining Association Rules

► Problem statement

Given

- a *minimum support* (min_sup), also called *support threshold*, and
- a *minimum confidence* (min_conf), also called *confidence threshold*,

find all association rules that satisfy both min_sup and min_conf from a data set D .

Basic Concepts

- ▶ **Strong rules:**

An association rule is *strong* if it satisfies both min_sup and min_conf .

- ▶ **k -itemset:** an itemset that contains k items.

- ▶ {computer, financial_management_software} is a 2-itemset.

- ▶ **Support count** of an itemset in data set D : number of transactions in D that contain the itemset.

- ▶ **Minimum support count** = $min_sup \times$ total number of transactions in a data set.

- ▶ **Frequent itemset** in a data set D : itemset whose support count in D is at least the minimum support count.

How to Mine Association Rules

- ▶ A two-step process:
 - ▶ Find all frequent itemsets ---- the key step
 - ▶ Generate strong association rules from frequent itemsets.
- ▶ Example: given min_sup=50% and min_conf=50%

Transaction ID	Items Bought
2000	A,B,C
1000	A,C
4000	A,D
5000	B,E,F



Frequent Itemset	Support
{A}	75%
{B}	50%
{C}	50%
{A, C}	50%

- ▶ Generate strong rules:
 - ▶ {A} → {C} [support=50%, confidence=66.6%]
 - ▶ {C} → {A} [support=50%, confidence=100%]